

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1

Answer the following:

- (a) Write the expressions for velocity in polar coordinates (v_r and v_θ components) and state when polar coordinates are preferred over Cartesian. [0.5]

Ans:

In polar coordinates:

$$v_r = \dot{r} \text{ (radial component)}$$

$$v_\theta = r\dot{\theta} \text{ (transverse component)}$$

$$\rightarrow \mathbf{v} = \dot{r} \hat{e}_r + r\dot{\theta} \hat{e}_\theta$$

Polar coordinates are preferred when the motion has radial symmetry or when dealing with central-force problems (e.g. planetary motion, pendulums), because the constraint is naturally expressed in r and θ .

- (b) Define the Lagrangian L of a mechanical system and state the Euler-Lagrange equation in terms of a generalized coordinate q . [0.5]

Ans:

The Lagrangian is defined as $L = T - V$, where T = kinetic energy and V = potential energy.

The Euler-Lagrange equation is:

$$d/dt (\partial L / \partial \dot{q}) - \partial L / \partial q = 0$$

This yields the equation(s) of motion for the system in terms of the generalized coordinate q .

- (c) State the transport theorem for a vector quantity as observed from a fixed (inertial) frame and a rotating frame. Identify the correction term. [0.5]

Ans:

The transport theorem states:

$$(d\mathbf{A}/dt)_{\text{fixed}} = (d\mathbf{A}/dt)_{\text{rotating}} + \boldsymbol{\Omega} \times \mathbf{A}$$

where $\boldsymbol{\Omega}$ is the angular velocity of the rotating frame.

The correction term $\boldsymbol{\Omega} \times \mathbf{A}$ accounts for the rotation of the basis vectors themselves as seen from the fixed frame.

- (d) Define the damping ratio ζ and state the three response regimes (underdamped, critically damped, overdamped) with their conditions. [0.5]

Ans:

The damping ratio $\zeta = c / c_{cr} = c / (2\sqrt{km})$, where c = damping coefficient, k = stiffness, m = mass.

Three regimes:

$\zeta < 1 \rightarrow$ Underdamped: oscillatory with exponentially decaying amplitude.

$\zeta = 1 \rightarrow$ Critically damped: fastest return to equilibrium without overshoot.

$\zeta > 1 \rightarrow$ Overdamped: slow, non-oscillatory return to equilibrium.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

- (a) A particle moves in a circle of radius $r = 2$ m at constant angular speed $\dot{\theta} = 3$ rad/s. Calculate the radial (centripetal) acceleration a_r and the transverse acceleration a_θ . [0.5]

Ans:

For circular motion at constant angular speed ($\dot{r} = 0$, $\ddot{r} = 0$, $\dot{\theta} = 3$):

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 2 \times 3^2 = -18 \text{ m/s}^2 \text{ (directed radially inward)}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 0 = 0 \text{ m/s}^2$$

The magnitude of centripetal acceleration is 18 m/s^2 .

- (b) Write the kinetic energy T and potential energy V for a simple pendulum of mass m and length L , using θ as the generalized coordinate. [0.5]

Ans:

For a simple pendulum with angle θ :

$$T = \frac{1}{2} m (L\dot{\theta})^2 = \frac{1}{2} m L^2 \dot{\theta}^2$$

$$V = -mgL \cos \theta \text{ (taking the pivot as the datum)}$$

or equivalently $V = mgL(1 - \cos \theta)$ with datum at the lowest point.

The Lagrangian is $L = T - V = \frac{1}{2}mL^2\dot{\theta}^2 + mgL \cos \theta$.

(c) Define a holonomic constraint and state how the number of degrees of freedom (DOF) is determined for a system of N particles with k holonomic constraints. [0.5]

Ans:

A holonomic constraint is one that can be expressed as an equation relating only coordinates (and possibly time), not velocities: $f(q_1, q_2, \dots, t) = 0$.

DOF = $3N - k$, where N = number of particles, k = number of independent holonomic constraints.

Example: A bead on a wire ($k = 2$ of 3 coordinates constrained $\rightarrow$ 1 DOF).

(d) A spring-mass system has $m = 4$ kg and $k = 256$ N/m. Calculate the natural frequency ω_n (rad/s) and the time period T of free oscillation. [0.5]

Ans:

$$\omega_n = \sqrt{k/m} = \sqrt{256/4} = \sqrt{64} = 8 \text{ rad/s.}$$

$$T = 2\pi/\omega_n = 2\pi/8 = \pi/4 \approx 0.785 \text{ s.}$$

Solve any TWO from Q2, Q3, Q4 [3 Marks each = 6 Marks]

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO1, CO2

A simple pendulum consists of a point mass $m = 1$ kg attached to a massless, inextensible string of length $L = 1$ m. The bob swings in a vertical plane under gravity ($g = 9.81 \text{ m/s}^2$). Using the angle θ from the vertical as the generalized coordinate, analyze the system using the Lagrangian method. Use the following relations: Lagrangian $L = T - V$; Euler-Lagrange equation: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$; for small angles: $\sin \theta \approx \theta$.

(a) State the number of degrees of freedom of this system and identify the generalized coordinate. [0.5]

Ans:

The pendulum has 1 degree of freedom (DOF = 1).

Generalized coordinate: θ — the angle of the string measured from the vertical.

(b) Write the position of the mass in Cartesian coordinates (x, y) in terms of θ and L . [0.5]

Ans:

Taking the pivot as the origin, with x horizontal and y vertically downward:

$$x = L \sin \theta = 1.0 \sin \theta$$

$$y = L \cos \theta = 1.0 \cos \theta$$

$$\text{Differentiating: } \dot{x} = L\dot{\theta} \cos \theta, \dot{y} = -L\dot{\theta} \sin \theta$$

(c) Derive the kinetic energy T in terms of $\theta, \dot{\theta}, m$, and L . [0.5]

Ans:

$$v^2 = \dot{x}^2 + \dot{y}^2 = L^2\dot{\theta}^2 \cos^2 \theta + L^2\dot{\theta}^2 \sin^2 \theta = L^2\dot{\theta}^2$$

$$T = \frac{1}{2}mv^2 = \frac{1}{2} \times 1.0 \times 1.0^2 \times \dot{\theta}^2 = \frac{1}{2}mL^2\dot{\theta}^2$$

$$T = \frac{1}{2}(1.0)(1.0^2)\dot{\theta}^2 = 0.5 \dot{\theta}^2 \text{ J}$$

(d) Write the potential energy V in terms of θ, m, g , and L . [0.5]

Ans:

Taking the pivot as the datum (y positive downward):

$$V = -mgy = -mgL \cos \theta = -1.0 \times 9.81 \times 1.0 \times \cos \theta$$

$$V = -9.81 \cos \theta \text{ J}$$

(e) Form the Lagrangian $L = T - V$ and apply the Euler-Lagrange equation to derive the equation of motion. [0.5]

Ans:

$$L = T - V = \frac{1}{2}mL^2\dot{\theta}^2 + mgL \cos \theta$$

$$\text{Euler-Lagrange: } d/dt(\partial L/\partial \dot{\theta}) - \partial L/\partial \theta = 0$$

$$\partial L/\partial \dot{\theta} = mL^2\dot{\theta} = 1.0 \times 1.0^2 \times \dot{\theta} = 1.0 \dot{\theta}$$

$$d/dt(mL^2\dot{\theta}) = mL^2\ddot{\theta} = 1.0 \ddot{\theta}$$

$$\partial L/\partial \theta = -mgL \sin \theta = -9.81 \sin \theta$$

$$\text{EOM: } 1.0 \ddot{\theta} + 9.81 \sin \theta = 0$$

$$\text{Simplifying (divide by } mL^2): \ddot{\theta} + (g/L) \sin \theta = 0$$

$$\rightarrow \ddot{\theta} + (9.81/1.0) \sin \theta = 0 \rightarrow \ddot{\theta} + 9.81 \sin \theta = 0$$

(f) For small oscillations ($\sin \theta \approx \theta$), determine the natural frequency of the pendulum. [0.5]

Ans:

$$\text{For small angles, } \sin \theta \approx \theta \rightarrow \ddot{\theta} + (g/L)\theta = 0$$

$$\text{This is SHM with } \omega_n^2 = g/L = 9.81/1.0 = 9.81$$

$$\omega_n = \sqrt{9.81} = 3.132 \text{ rad/s}$$

$$\text{Time period } T = 2\pi/\omega_n = 2\pi/3.132 = 2.006 \text{ s}$$

Q3 [3 Marks]

Bloom: BL 3–4 (Apply / Analyze) | CO2

A horizontal platform rotates at a constant angular velocity $\Omega = 4 \text{ rad/s}$ about a vertical axis. An object at a radial distance $r = 1.5 \text{ m}$ from the center moves radially outward at a constant speed $v_{\text{rel}} = 2 \text{ m/s}$ ($\dot{r} = 2 \text{ m/s}$, $\ddot{r} = 0$). The angular acceleration of the platform is zero ($\alpha = 0$).

- (a) Write the general expression relating the acceleration observed from the fixed (inertial) frame to quantities measured in the rotating frame. Identify each term. [0.5]

Ans:

The fixed-frame (inertial) acceleration, expressed in terms of rotating-frame quantities, is:

$$\mathbf{a}_{\text{fixed}} = \mathbf{a}_{\text{rel}} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$$

where:

$\mathbf{a}_{\text{rel}}$ = acceleration of the object in the rotating frame ($\ddot{r} = 0$ here),

$2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}}$ = Coriolis acceleration,

$\boldsymbol{\alpha} \times \mathbf{r}$ = Euler (angular acceleration) term (zero here since $\alpha = 0$),

$\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ = centripetal acceleration (directed radially inward).

- (b) Calculate the centripetal acceleration magnitude at $r = 1.5 \text{ m}$. [0.5]

Ans:

$$a_{\text{centripetal}} = \Omega^2 r = 4.0^2 \times 1.5 = 16 \times 1.5 = 24.0 \text{ m/s}^2$$

Direction: radially inward (toward the axis of rotation).

- (c) Calculate the Coriolis acceleration magnitude. [0.5]

Ans:

$$a_{\text{Coriolis}} = 2\Omega v_{\text{rel}} = 2 \times 4.0 \times 2.0 = 16.0 \text{ m/s}^2$$

- (d) State the direction of the Coriolis acceleration relative to the radial velocity. [0.5]

Ans:

By the right-hand rule for $2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}}$ ($\boldsymbol{\Omega}$ along vertical, $\mathbf{v}_{\text{rel}}$ radially outward): the Coriolis acceleration is perpendicular to $\mathbf{v}_{\text{rel}}$, directed tangentially in the direction opposite to the platform's rotation.

- (e) Calculate the total acceleration magnitude of the object as observed from the fixed (inertial) frame. [0.5]

Ans:

$$\begin{aligned} |a_{\text{total}}| &= \sqrt{a_{\text{cent}}^2 + a_{\text{Cor}}^2} = \sqrt{(24.0)^2 + (16.0)^2} \\ &= \sqrt{576 + 256} = \sqrt{832} = 28.84 \text{ m/s}^2 \end{aligned}$$

- (f) If the object continues at the same radial speed, find the centripetal acceleration when $r = 3 \text{ m}$ and comment on how it changes. [0.5]

Ans:

$$\text{At } r = 3 \text{ m: } a_{\text{centripetal}} = \Omega^2 r = 4.0^2 \times 3 = 48.0 \text{ m/s}^2$$

Compared to 24.0 m/s^2 at $r = 1.5 \text{ m}$, the centripetal acceleration has doubled. It increases linearly with radial distance ($a_{\text{cent}} \propto r$), while the Coriolis acceleration (16.0 m/s^2) remains unchanged since v_{rel} is constant.

Q4 [3 Marks]

Bloom: BL 4–5 (Analyze / Evaluate) | CO2, CO3

A mass $m = 3 \text{ kg}$ is attached to a spring of stiffness $k = 300 \text{ N/m}$ and a viscous damper with damping coefficient $c = 12 \text{ N}\cdot\text{s/m}$. The system undergoes free vibration after being displaced 0.05 m from equilibrium and released from rest. Take the equilibrium position as the origin.

- (a) Write the equation of motion for this damped spring-mass system. [0.5]

Ans:

The equation of motion for a damped SDOF system:

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$3.0\ddot{x} + 12.0\dot{x} + 300.0x = 0$$

$$\text{Dividing by } m: \ddot{x} + 4.0\dot{x} + 100.0x = 0$$

$$\text{Or: } \ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0$$

- (b) Calculate the natural frequency ω_n and the critical damping coefficient c_{cr} . [0.5]

Ans:

$$\omega_n = \sqrt{k/m} = \sqrt{300.0/3.0} = \sqrt{100} = 10.00 \text{ rad/s}$$

$$c_{\text{cr}} = 2\sqrt{km} = 2\sqrt{300.0 \times 3.0} = 2\sqrt{900} = 2 \times 30.00 = 60.00 \text{ N}\cdot\text{s/m}$$

- (c) Calculate the damping ratio ζ and state the type of damping. [0.5]

Ans:

$$\zeta = c / c_{\text{cr}} = 12.0 / 60.00 = 0.2000$$

Since $\zeta = 0.2000 < 1 \rightarrow$ the system is UNDERDAMPED.

It will oscillate with exponentially decaying amplitude.

(d) Calculate the damped natural frequency ω_d . [0.5]

Ans:

$$\begin{aligned}\omega_d &= \omega_n \sqrt{1 - \zeta^2} = 10.00 \times \sqrt{1 - 0.2000^2} \\ &= 10.00 \times \sqrt{1 - 0.040000} \\ &= 10.00 \times 0.979796 \\ &= 9.7980 \text{ rad/s} \approx 9.80 \text{ rad/s}\end{aligned}$$

(e) Calculate the logarithmic decrement δ and find the ratio of two successive peak amplitudes. [0.5]

Ans:

$$\begin{aligned}\delta &= 2\pi\zeta / \sqrt{1 - \zeta^2} = 2\pi \times 0.2000 / \sqrt{1 - 0.040000} \\ &= 1.2566 / 0.979796 \\ &= 1.2825\end{aligned}$$

$$\text{Ratio of successive peaks: } x_n / x_{(n+1)} = e^\delta = e^{1.2825} = 3.6058$$

$$\text{Or equivalently } x_{(n+1)} / x_n = e^{(-\delta)} = 0.2773$$

Each successive peak is $\approx 27.73\%$ of the previous one.

(f) If the damper is removed ($c = 0$), write the displacement as a function of time $x(t)$ for the given initial conditions. [0.5]

Ans:

With $c = 0$ (undamped), EOM: $m\ddot{x} + kx = 0$

$$x(t) = x_0 \cos(\omega_n t) = 0.05 \cos(10.00 t) \text{ m}$$

(Since initial conditions are $x(0) = 0.05 \text{ m}$, $\dot{x}(0) = 0$, the solution is a pure cosine with amplitude 0.05 m and period $T = 2\pi/10.00 = 0.6283 \text{ s}$)

— End of Set 1 Model Answers —